Dynamic Valuation of a Soccer Club

Discounted Cash Flows and the Bellman Equation

Nico Rosamilia
Sport Finance and Economics

1 Economic question and connection to the previous model

In the optimal talent spending model, clubs chose talent to maximize one-season profit. Here the question changes: *What is a club worth today when its talent survives across seasons and generates future cash flows?*

Retain the sporting and revenue functions already introduced in class:

$$s(x, x_j) = \frac{x}{x + x_j}, \qquad x_j > 0, \tag{1}$$

$$R(x, x_j) = \bar{R} + m\left[a\, s(x, x_j) - \frac{b}{2} s(x, x_j)^2\right], \qquad a > b > 0, \quad m > 0. \tag{2}$$

The rival's talent x_j is held constant. We take the previously derived marginal-revenue expression as known:

$$R_x(x, x_j) = m\,(a - b\, s(x, x_j))\,\frac{x_j}{(x + x_j)^2}. \tag{3}$$

In this lecture, x denotes a *stock of productive talent*, rather than a monetary expenditure. The transfer price q converts talent units into money.

2 Discounted cash flows and the talent stock

A euro received next season is worth less than a euro received today. With an annual discount rate $r > 0$ and discount factor $\beta = (1 + r)^{-1}$,

$$PV = \sum_{t=1}^{\infty} \beta^t CF_t. \tag{4}$$

If net cash flow is the same each year, $CF_t = CF$, then

$$PV = CF \sum_{t=1}^{\infty} \beta^t = CF \frac{\beta}{1 - \beta} = \boxed{\frac{CF}{r}}. \tag{5}$$

Let y be the productive talent stock from the preceding season, observed immediately before the current transfer decision. A fraction $1 - \delta$ of it remains usable ($0 < \delta < 1$). The club chooses current talent $x \geq 0$ by trading

$$\tau = x - (1 - \delta)y. \tag{6}$$

A positive τ is a net purchase; a negative τ is a net sale. The club pays $q\tau$ *at the beginning* of the season. At the *end* of the season it receives operating cash flow

$$R(x, x_j) - f - wx, \tag{7}$$

where f is a fixed operating cash cost and w is annual wages per unit of talent. Next season's inherited talent state is $y' = x$; depreciation applies before next season's transfer decision.

Assume constant q, w, r, δ, frictionless buying and selling, no financing constraint, and a stationary rival stock. All monetary amounts are in the same units.

3 The Bellman equation

A *Bellman equation* expresses the maximum value of a decision problem as the best current payoff plus the discounted value of decisions that remain in the future. It solves a multi-period problem recursively.

Let $V(y)$ be the value of the club at the beginning of a season, before transfers, given inherited talent y. Its problem is

$$\boxed{V(y) = \max_{x \geq 0} \left\{ -q[x - (1-\delta)y] + \beta[R(x, x_j) - f - wx + V(x)] \right\}.} \tag{8}$$

The term $-q[x - (1-\delta)y]$ is the immediate transfer cash flow; $R - f - wx$ is the end-of-season operating cash flow; and $V(x)$ is the value of continuing into the next season. Both end-of-season terms are discounted by β.

4 Solve the Bellman equation step by step

Step 1. Conjecture the form of the value function

Because an additional unit of inherited talent can be traded at a constant price, try an affine value function:

$$\boxed{V(y) = A + By}, \tag{9}$$

where A and B are unknown constants. In the continuation term, $V(x) = A + Bx$.

Step 2. Substitute the conjecture

Insert (9) into (8):

$$A + By = \max_{x \geq 0} \left\{ -qx + q(1-\delta)y + \beta[R(x, x_j) - f - wx + A + Bx] \right\} \tag{10}$$

$$= q(1-\delta)y + \beta A + \max_{x \geq 0} \left\{ \beta[R(x, x_j) - f - wx] + (-q + \beta B)x \right\}. \tag{11}$$

The maximized expression does not contain y. Equality for every inherited stock y therefore requires

$$\boxed{B = q(1-\delta)}. \tag{12}$$

Interpretation: one more unit of inherited talent adds its remaining resale value to club value.

Step 3. Reduce the dynamic choice to an annual surplus problem

Replace B in (11):

$$A = \beta A + \max_{x \geq 0} \{\beta[R(x, x_j) - f - wx] - qx + \beta q(1-\delta)x\} \tag{13}$$

$$= \beta A + \beta \max_{x \geq 0} \left\{ R(x, x_j) - f - \left[w + \frac{q}{\beta} - q(1-\delta) \right] x \right\}. \tag{14}$$

Since $1/\beta = 1 + r$,

$$w + \frac{q}{\beta} - q(1-\delta) = w + q(1+r) - q(1-\delta) = \boxed{w + q(r+\delta)}. \tag{15}$$

Define the *annual economic user cost of talent* as

$$\boxed{c = w + qr + q\delta}. \tag{16}$$

It contains annual wages w, the opportunity cost of transfer capital qr, and economic depreciation $q\delta$. Equation (14) becomes

$$A = \beta A + \beta \max_{x \geq 0} \{R(x, x_j) - f - cx\}. \tag{17}$$

Step 4. Identify the optimal talent policy

Define

$$x^* \in \arg\max_{x \geq 0} \{R(x, x_j) - f - cx\}. \tag{18}$$

This is precisely the *form* of the one-season talent choice already studied, with the marginal cost c now derived from discounting and depreciation. For an interior solution, the first-order condition is

$$R_x(x^*, x_j) = c, \tag{19}$$

$$m \left(a - b \frac{x^*}{x^* + x_j} \right) \frac{x_j}{(x^* + x_j)^2} = \boxed{w + q(r+\delta)}. \tag{20}$$

For $x_j > 0$ and $a > b > 0$, revenue is strictly concave in own talent. In particular,

$$R_{xx}(x, x_j) = -m \left[\frac{b x_j^2}{(x + x_j)^4} + \frac{2(a - bs(x, x_j))x_j}{(x + x_j)^3} \right] < 0. \tag{21}$$

Thus (20) identifies the unique interior optimum when $c < ma/x_j$; otherwise the optimum is $x^* = 0$. We *reuse* the solution of the earlier talent problem rather than repeat its full derivation.

The optimal policy is stationary: $x_t = x^*$ every season. Net transfers are

$$\boxed{\tau_0 = x^* - (1-\delta)y, \qquad \tau_t = \delta x^* \quad (t \geq 1).} \tag{22}$$

Step 5. Solve for the remaining coefficient A

Write the maximum annual economic surplus as

$$M = R(x^*, x_j) - f - cx^*. \tag{23}$$

Then (17) gives

$$A = \beta A + \beta M, \tag{24}$$

$$(1-\beta)A = \beta M, \tag{25}$$

$$A = \frac{\beta}{1-\beta} M = \frac{1}{r} M. \tag{26}$$

Hence

$$\boxed{A = \frac{R(x^*, x_j) - f - cx^*}{r}}. \tag{27}$$

Step 6. Obtain club value

Combining (9), (12), and (27),

$$\boxed{V(y) = \underbrace{q(1-\delta)y}_{\text{value of usable inherited talent}} + \underbrace{\frac{R(x^*, x_j) - f - [w + q(r+\delta)]x^*}{r}}_{\text{present value of annual economic surplus}}}. \tag{28}$$

This formula is the solution of the Bellman equation under the assumptions above. It does not add a squad appraisal to an independently calculated club DCF: both terms arise from one consistent cash-flow problem.

5 Numerical application: use the previous talent solution

Use the parameters from the previous optimal talent spending lecture (in hypothetical monetary and talent units):

$$m = 100, \quad a = 2, \quad b = 1, \quad x_j = 10, \quad \bar{R} = f = 0, \quad c = 2. \tag{29}$$

The earlier first-order condition was

$$100(2-s)\frac{10}{(x+10)^2} = 2, \qquad s = \frac{x}{x+10}. \tag{30}$$

Because $x = 10s/(1-s)$ and $x + 10 = 10/(1-s)$, it reduces to

$$(2-s)(1-s)^2 = 0.2. \tag{31}$$

Its economically relevant solution is

$$s^* \approx 0.6193905, \qquad x^* = \frac{10s^*}{1-s^*} \approx 16.27365. \tag{32}$$

Now introduce the *new* dynamic valuation parameters:

$$w = 1, \qquad q = 5, \qquad r = 0.10, \qquad \delta = 0.10, \qquad y = 10. \tag{33}$$

They preserve the previous marginal cost:

$$c = 1 + 5(0.10 + 0.10) = 2. \tag{34}$$

Annual revenue at the inherited solution is

$$R^* = 100\left(2s^* - \frac{(s^*)^2}{2}\right) \approx 104.69588. \tag{35}$$

Therefore

$$M = R^* - cx^* \approx 104.69588 - 2(16.27365) \approx 72.14857, \tag{36}$$

$$q(1-\delta)y = 5(0.90)(10) = 45. \tag{37}$$

Apply (28):

$$\boxed{V(10) = 45 + \frac{72.14857}{0.10} \approx 766.48575.} \tag{38}$$

6 Check the answer using a direct DCF

This second method verifies the Bellman solution using actual transfer and operating cash flows rather than economic user cost.

At date 0, the club must acquire

$$\tau_0 = x^* - (1-\delta)y = 16.27365 - 9 = 7.27365 \tag{39}$$

units of talent. Its immediate net transfer cash flow is approximately

$$-q\tau_0 = -5(7.27365) = -36.36825. \tag{40}$$

In each subsequent year, the club receives revenue, pays wages, and purchases δx^* units of replacement talent. The recurring end-of-year net cash flow is

$$CF = R^* - f - wx^* - q\delta x^* \tag{41}$$

$$\approx 104.69588 - 16.27365 - 5(0.10)(16.27365) \tag{42}$$

$$\approx 80.28540. \tag{43}$$

The direct DCF is therefore

$$\boxed{V(10) = -36.36825 + \frac{80.28540}{0.10} \approx 766.48575.} \tag{44}$$

The equality can also be established algebraically:

$$V(y) = -q[x^* - (1-\delta)y] + \frac{R^* - f - (w + q\delta)x^*}{r} \tag{45}$$

$$= q(1-\delta)y + \frac{R^* - f - [w + q(r+\delta)]x^*}{r}. \tag{46}$$

This reproduces (28). There is no double counting of transfer payments: the date-0 acquisition and later replacement purchases are separate cash flows.

Main result

The Bellman equation shows how a discounted, infinite-horizon investment problem can be written recursively and solved by guessing the form of its value function. In this frictionless, stationary model, the optimal talent rule is the previously studied marginal-revenue condition with the *dynamic* user cost $w + q(r + \delta)$. Club value is the inherited usable talent value plus the present value of optimal annual economic surplus.